

Mahanidhi G K

AI & Machine Learning Student | Full-Stack Web Developer
| Deep Learning Enthusiast

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Education

B.Tech in Computer Science Engineering
Specialization: Artificial Intelligence & Machine Learning
Jain University, Bangalore
CGPA: 8.7

2023 - 2027 (Expected)

Experience

Mt. Blue Properties
Web Development Intern

Jun 2025 - Jul 2025

- Engineered a social media backend using 'Phlame' (custom PHP micro-framework) and built a Python-based CLI frontend.
- Gained hands-on experience in cross-language integration and API-centric system design.

Grazign
Automation Research Intern

July 2025 - Aug 2025

- Designed an ML object detection pipeline to automate property transcriptions and reduce manual errors.
- Developed workflows and virtual assistance strategies to optimize B2B client services and streamline operations.

Projects

Multi-Modal AI for Vehicle Damage and Claim Assessment | *Tech: TensorFlow, Scikit-learn, Streamlit, Pandas, Plotly*

- An end-to-end AI-powered Streamlit web application with interactive Plotly visualizations to automate vehicle damage detection and monetary claim estimation.
- Trained a deep learning image classification model using EfficientNetB0 Transfer Learning to identify vehicle damage and estimate severity from user-uploaded photos.
- Developed a CatBoost-based algorithm ($R^2 = 0.8058$) and a complete data pipeline to process features and predict specific amounts of monetary claims.

Applied Data Analysis Projects:

- COVID-19 Economic Impact Analysis:** Created a global trend dashboard with 10+ visualizations by building an ETL pipeline that merged COVID-19 and World Bank macroeconomic data from across 190+ countries.
- Video Games Sales & Review Analysis:** Processed a 60,000+ record Steam dataset using a dynamic cleaning pipeline, engineering custom KPIs to produce 10+ visualizations.
- Multi-Market Risk & Volatility analysis (Stock Market):** Ingested 3 indices (SP 500, NASDAQ, NIFTY 50) and 7 U.S./Indian large-cap equities via Yahoo Finance API to build a modular framework for multi-asset performance and trend visualization.

Document De-Jargonizer & Simplification Engine | *Tech: React, Python (FastAPI/Flask), Google Gemini API, Hugging Face, Tesseract OCR, Firebase*

- Engineered a full-stack AI system using Google Gemini 2.5-flash and Hugging Face BART to generate structured summaries and risk assessments from complex documents, featuring a 30+ language translation service via deep-translator.
- Architected a secure Python backend with Firebase Firestore and JWT authentication, integrating a robust text extraction pipeline utilizing Tesseract OCR and pdfplumber for accurate data ingestion.

USD-INR Time Series Forecasting Model | *Tech: Python, Statsmodels, Pandas, Yahoo Finance API*

- Built a robust ARIMA time-series forecasting model to predict future USD/INR exchange rates (2026-2027), utilizing the Yahoo Finance API for data acquisition and ADF testing for time-series stationarity.
- Optimized the model architecture via AIC-based hyperparameter tuning and validated predictive reliability using walk-forward validation and standard error metrics (MSE, RMSE, R^2).

Technical Skills & Interests

Programming Languages: Python, Java, SQL, HTML, CSS, PHP

Libraries and Frameworks: Pandas, Numpy, Matplotlib, Seaborn, Plotly Express, PyTorch, TensorFlow, ScikitLearn

Developer Tools: VS Code, Git, Github, Tableau, Docker

Relevant Coursework: Data Structures and Algorithms, Operating System, Computer Networks, DBMS, Data Analytics, Machine Learning, Deep Learning (Neural Networks), Computer Vision

Achievements

- Won third place in the Google TechSprint Hackathon conducted by GDG.
- Selected in the inter-college round of Smart India Hackathon (2025) and successfully developed and submitted 2 distinct projects across different domains.